What is ecological engineering?

Ecological engineering, defined as the design of sustainable ecosystems that integrate human society with **its** natural environment for the benefit of both, has developed over the last 30 years, and rapidly over the last 10 years. **Its** goals include the restoration of ecosystems that have been substantially disturbed by human activities and the development of new sustainable ecosystems that have both human and ecological values. It is especially needed as conventional energy sources diminish and amplification of nature's ecosystem services is needed even more. There are now several universities developing academic programs or departments called ecological engineering, ecological restoration, or similar terms, the number of manuscripts submitted to the journal Ecological Engineering continue to increase at a rapid rate, and the U.S. National Science Foundation now has a specific research focus area called ecological engineering. There are many private firms now developing and even prospering that are now specializing in the restoration of streams, rivers, lakes, forests, grasslands, and wetlands, the rehabilitation of minelands and urban brownfields, and the creation of treatment wetlands and phytoremediation sites. It appears that the perfect synchronization of academy, publishing, research resources, and practice is beginning to develop. Yet the field still does not have a formal accreditation in engineering and receives guarded acceptance in the university system and workplace alike.

Some years ago, **I** presented a paper at the first EcoSummit in Copenhagen, Denmark, that provided several recommendations for the then newly forming field of ecological engineering. The paper was titled the "Ecological engineering—the 7-year itch" because it was presented seven years after our definition and principles were published in the first textbook on ecological engineering entitled "Ecological Engineering: An Introduction to Ecotechnology". I was starting to get nervous that the field was straying from principles and to a hodge-podge of case studies and pollution control projects, many of **which** were not even ecological engineering. I was also concerned that ecological engineering was not being embraced by either the engineering or ecology communities and that the convergence of **these two fields** into a third transdisciplinary field was simply not happening. In that paper, I presented six recommendations:

1) Ecologists need to recognize the applied nature of **their** field to offer prescriptions, not just descriptions, for environmental problems. 2)Engineers need to understand that biological and ecological sciences are fundamental to their tasks. 3) A formal accreditation of ecological engineering should be developed in concert with existing engineering accreditations. 4)Universities need to integrate ecology and engineering into rational and rigorous programs of ecological engineering. 5)Ecologists and engineers need to work together and understand each other's language. 6) An international dialog needs to continue to establish the scientific basis, limitations and opportunities for ecological engineering.

This paper will review the history of ecological engineering and **its** progress in light of these 16-year-old recommendations. In that period, the world has changed and ecological engineering may be more relevant than ever in times of diminishing resources and calls for sustainability.

The fundamental unit of ecological engineering must remain the ecosystem—ecological engineering must involve the creation and restoration of an ecosystem. An ecosystem's presence or absence is the fundamental test of whether someone is talking about ecological engineering, or perhaps only environmental engineering (the application of technology to solve environmental problems; principle examples include design of wastewater treatment, water treatment, and air pollution control systems) or simply "green engineering" (sometimes referred to as ecotechniques, or even green architecture). Natural ecosystems are complex entropy-fighting systems, and in that complexity comes an infinite amount of feedbacks and adaptations that contribute to resiliency. Human society, as a "part of" nature and not "apart from" nature would do well to recognize and use the important functions of nature to provide a resilient and sustainable society.

Nombre:……legajo:……curso:……..

1) Responder las siguientes preguntas en ESPAÑOL.

   - ¿En qué se están especializando muchas firmas privadas?

2) Traducir los siguientes fragmentos al ESPAÑOL.

   - This paper will review the history of ecological engineering

   - Ecological engineering must involve the creation and restoration of an ecosystem.

3) Referencia contextual. Indicar a qué o quién hacen referencia las siguientes palabras o frases.

   - its (línea 2) : _____
   - its (línea 4) : _____
   - I (línea 16) : _____
   - which (línea 21) : _____
   - These two fields (línea 23) _____
   - Their (línea 25) : _____
   - its (línea 32) : _____

4) Indicar los antónimos de las siguientes palabras
   - apart (línea 29) _____
   - artificial (línea 3) _____

5) Indicar los sinónimos de las siguientes palabras
   - suggestions (línea 24) _____
   - traditional (línea 6) _____
   - Restrictions (línea 31) _____

6) Completar el siguiente cuadro de conectores en ESPAÑOL.

CONECTOR	TRADUCCIÓN	TIPO DE RELACIÓN	IDEA UNO	IDEA DOS
Both…and (línea 5)				
Yet (línea 14)				
Whether - ….or (línea 37)				
to (línea 43)				

RESUELTO

Ejemplo de parcial · Ecological engineering

Las respuestas van en azul. Las consignas son las mismas de las hojas anteriores; los números entre paréntesis son los renglones del texto.

1) Responder las siguientes preguntas en ESPAÑOL.

• ¿En qué se están especializando muchas firmas privadas?
→ En la restauración de arroyos, ríos, lagos, bosques, pastizales y humedales; en la rehabilitación de tierras mineras y de terrenos urbanos abandonados o contaminados (brownfields), y en la creación de humedales de tratamiento y sitios de fitorremediación (renglones 10 a 13).

2) Traducir los siguientes fragmentos al ESPAÑOL.

• This paper will review the history of ecological engineering
→ Este trabajo repasará (revisará) la historia de la ingeniería ecológica.

• Ecological engineering must involve the creation and restoration of an ecosystem.
→ La ingeniería ecológica debe implicar la creación y la restauración de un ecosistema.

3) Referencia contextual. Indicar a qué o quién hacen referencia las siguientes palabras o frases.

• its (línea 2)
→ human society: la sociedad humana (su ambiente natural)

• its (línea 4)
→ ecological engineering: la ingeniería ecológica (sus objetivos)

• I (línea 16)
→ el autor (referencia exofórica: primera persona)

• which (línea 21)
→ case studies and pollution control projects: los estudios de caso y los proyectos de control de la contaminación

• These two fields (línea 23)
→ engineering and ecology: la ingeniería y la ecología

• Their (línea 25)
→ ecologists: los ecólogos

• its (línea 32)
→ ecological engineering: la ingeniería ecológica (su progreso)

4) Indicar los antónimos de las siguientes palabras.

• apart (línea 29)
→ together («Ecologists and engineers need to work together»)

• artificial (línea 3)
→ natural («natural environment»)

5) Indicar los sinónimos de las siguientes palabras.

• suggestions (línea 24)
→ recommendations

• traditional (línea 6)
→ conventional

• Restrictions (línea 31)
→ limitations

6) Completar el siguiente cuadro de conectores en ESPAÑOL.

Conector	Traducción	Tipo de relación	Idea uno	Idea dos
Both…and (línea 5)	tanto… como	adición	Los nuevos ecosistemas sustentables tienen valores humanos…	…y también valores ecológicos.
Yet (línea 14)	sin embargo	contraste	Parece que empieza a darse la sincronización perfecta entre la academia, las publicaciones, los recursos de investigación y la práctica.	El campo todavía no tiene una acreditación formal en ingeniería y lo aceptan con reservas en la universidad y en el trabajo.
Whether… or (línea 37)	si… o	disyunción (alternancia)	Alguien habla de ingeniería ecológica…	…o quizás solo de ingeniería ambiental (o simplemente de «ingeniería verde»).
to (línea 43)	para	propósito (finalidad)	La sociedad haría bien en reconocer y usar las funciones importantes de la naturaleza…	…para lograr una sociedad resiliente y sustentable.